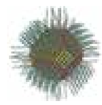
World's fastest spinning object
It is a dumbbell-shaped silica nanoparticle object which spins at 300 bn revolutions per minute. **https://dgit.in/feb20-31**



Tiny crystals generate heat
Researchers found that nanocrystals with the right kind of imperfections can generate heat. **https://dgit.in/feb20-32**

CRYSTAL SYSTEMS

Nature is often perceived to consist of largely organic shapes and curves but the geometric world of crystals have a different story to tell

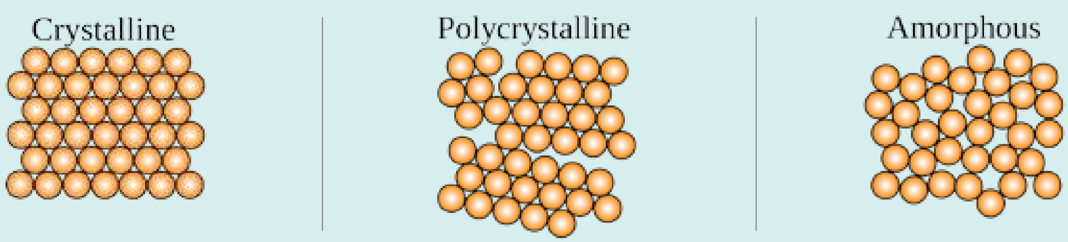
Mithun Mohandas | mithun@digit.in

If you were asked to think about the epitome of crystalline beauty, diamonds were probably the first thing to come to mind. It's well known that diamonds and carbon dust are practically the same thing but the two appear vastly different thanks to their different crystalline structures.

Not just diamonds but a lot of precious and semi-precious gems look the way they do because of the way that the constituent elements – which can be atoms, molecules or even ions – are arranged within them.

Geodes are a great example of how crystals are formed as you can see an amorphous structure slowly transforming to develop crystalline features. And between the two you have the polycrystalline formations.

Put separately, you can see how the different structural arrangement of atoms can affect how a substance appears. On the left, we have Silicon in a pure crystalline form which appears to be a shiny metallic cylinder with a slight blue-ish tinge. In the centre, we have a polycrystalline silicon which can only be produced using a chemical purification process. It doesn't feature a crystalline appearance because of the method used to produce it but it is very pure. And lastly, we have the amorphous silicon which appears to be in a powder form because of the impurities present. Depending on which impurities it has, the colour of the amorphous form can vary. One of the most stable and abundant forms of amorphous silicon would be Silica (Silicon Dioxide) which is white in colour.



Credit: Wikimedia